CASH MANAGEMENT

1. Cash Management

- Model Based
  - William Baumol
  - Miller-Orr Model
- Cash Budget
  - Cash Plan
    - Inflow
    - Outflow
    - Surplus / Deficit
      - In Case of Surplus
        - Invest or Repay of existing Borrowing
      - In Case of Deficit
        - Borrow or Sale of exist. investment

2. William Baumol Model

If low cash balance is maintained – transaction cost will be high

If high cash balance is maintained – Opportunity cost will be high

Optimum Cash Balance – where total of transaction cost & opportunity cost is minimum

$$\boxed{\text{Optimum Cash balance} = \sqrt{\frac{2 \times U \times P}{S}}}$$

where, U = Annual cash requirement

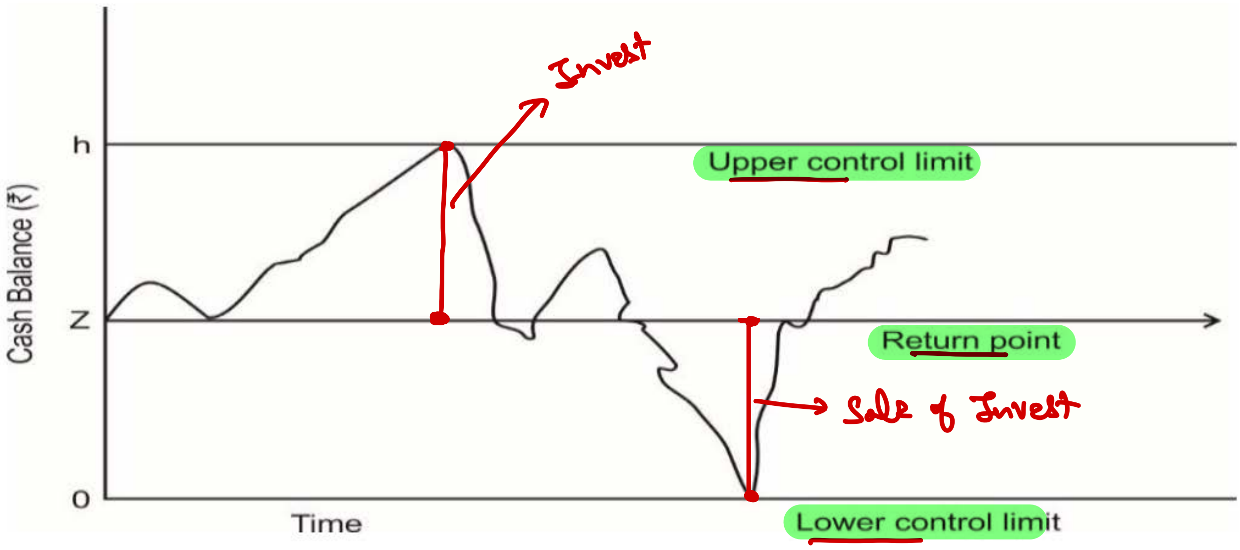
P = Fixed cost per transaction

S = opportunity cost for ₹ 1 p.a.

3. Miller-Orr Cash Management Model

This model is designed to determine the time and size of transfers between an investment account and cash account. In this model control limits are set for cash balances. These limits may consist of h as upper limit, z as the return point and zero as the lower limit.

- When the cash balance reaches the upper limit, the transfer of cash equal to (h – z) is invested in marketable securities account.
- When it touches the lower limit, a transfer from marketable securities account to cash account is made.
- During the period when cash balance stays between (h,z) and (z,0) i.e. high and low limits no transactions between cash and marketable securities account is made.



Question – 1	$U = 90 \times 4 = 360$

K Ltd. has a Quarterly cash outflow of ₹ 9,00,000 arising uniformly during the Quarter. The company has an Investment portfolio of Marketable Securities. It plans to meet the demands for cash by periodically selling marketable securities. The marketable securities are generating a return of 12% p.a. Transaction cost of converting investments to cash is ₹ 60. The company uses Baumol model to find out the optimal transaction size for converting marketable securities into cash.
Consider 360 days in a year.

$S = \frac{12}{100} \times 1 = 0.12$

$\text{optimum CB} = \sqrt{\frac{2 \times 360 \times 60}{0.12}} = 60,000$

You are required to calculate:

(a) Company's average cash balance → $\frac{60000}{2} = 30000$

(b) Number of conversions each year and → $\frac{360}{60000} = 60$ ✓

(c) Time interval between two conversions

→ $\frac{360}{60} = 6$ days

Solution

(a) Annual cash outflows (U) = 9,00,000 × 4 = ₹ 36,00,000

Fixed cost per transaction (P) = ₹ 60

Opportunity cost of one rupee p.a. (S) = $\frac{12}{100} \times 1 = 0.12$

Optimum cash balance = $\sqrt{\frac{2 \times U \times P}{S}} = \sqrt{\frac{2 \times 36,00,000 \times 60}{0.12}}$ = ₹ 60,000

Average cash balance = $\frac{60,000}{2}$ = ₹ 30,000

(b) Number of conversions p.a. = $\frac{\textit{Annual requirement}}{\textit{Optimum cash balance}} = \frac{36,00,000}{60,000} = 60$

(c) Time interval between two conversion = $\frac{360}{\textit{No. of conversions}} = \frac{360}{60}$ = 6 days

4. Cash Budget

Performa Cash Budget

Particulars	April	May	June
Opening Balance (A)	-	-	-
Receipt:			
Sales realization	-	-	-
Advance from sale of assets or investment	-	-	-
Dividend	-	-	-
Total Receipt (B)	-	-	-
Payments:			
Creditors payment	-	-	-
Wages Payment	-	-	-
Overheads Payment	-	-	-
Payment related to purchase of assets	-	-	-
Dividend payment	-	-	-
Income tax	-	-	-
Total Payments (C)	-	-	-
Surplus/Deficit (A + B – C)	-	-	-
(-) Investment	-	-	-
(+) Sale of Investment	-	-	-
(+) Borrowing	-	-	-
(-) Repayment of Borrowing	-	-	-
Closing Balance	-	-	-

Invest / Finance

Question – 2	

A garment trader is preparing cash forecast for first three months of calendar year 2021. His estimated sales for the forecasted periods are as below:

	January (₹ '000)	February (₹ '000)	March (₹ '000)
Total Sales	600	600	800

(i) The trader sells directly to public against cash payments and to other entities on credit. Credit sales are expected to be four times the value of direct sales to public. He expects 15% customers to pay in the month in which credit sales are made, 25% to pay in the next month and 58% to pay in the next to next month. The outstanding balance is expected to be written off.

(ii) Purchase of goods are made in the month prior to sales and it amounts to 90% of sales and are made on credit. Payments of these occur in the month after the purchase. No inventories of goods are held.

(iii) Cash balance as on 1st January, 2021 is ₹ 50,000.

(iv) Actual sales for the last two months of calendar year 2020 are as below:

	November (₹ '000)	December (₹ '000)
Total Sales	640	880

You are required to prepare a monthly cash budget for the three months from January to March, 2021.

Solution

Given, Cash sales = 25% of credit sales

Thus, let credit sales = y ∴ Cash sales = 0.25y

∴ y + 0.25y = Total sales

1.25y = Total sales

$$y = \frac{Total\ Saels}{1.25}$$

y = 80% of total sales

Thus, Credit sales = 80% of total sales and Cash sales = 20% of total sales

Cash Budget

Particulars	Jan.	Feb.	March
Opening Balance (A)	**50**	**174.96**	**355.28**
Receipts			
20% of current month	120	120	160
12% of current month	72	72	96
20% of previous month	176	120	120
46.4% of previous to previous month	296.96	408.32	278.40
Total receipts (B)	**664.96**	**720.32**	**654.40**
Payments			
Creditors payment (90% x Sale of Co. Month)	540	540	720
Total payments (C)	**540**	**540**	**720**
Closing Balance (A + B - C)	**174.96**	**355.28**	**289.68**

Question – 3	

Following information relates to ABC Company for the year 2016:

(a) Projected Sales: (in ₹ lakhs)

Month	August	September	October	November	December
Sales	35	40	40	45	46

(b) Gross Profit Margin will be 20% on sales

(c) 10% of projected sales will be cash sales. Out of credit sales of each month, 50% will be collected in the next month and the balance will be collected during the second month following the month of sale.

(d) Creditors will be paid in the first month following credit purchase. There will be credit purchase only

o/s 1 Month

(e) Wages and salaries will be paid on the first day of the next month. The amount will be ₹ 3 lakhs each month.

(f) Interim dividend of ₹ 2 lakhs will be paid in December 2016.

(g) Machinery costing ₹ 10 lakhs will be purchased in September 2016. Repayment by instalment of ₹ 50,000 p.m., will start from October 2016.

(h) Administrative Expenses of ₹ 1,00,000 per month will be paid in the month of their incurrence.

(i) Assume no minimum cash balance is required. Opening cash balance as on 01.10.2016 is estimated at ₹ 10 lakhs.

Prepare the monthly cash budget for the 3 month period (October 2016 to December 2016).

Solution

Cash Budget for months from October to December (₹ lakhs)

	Particulars	October	November	December
A.	Opening balance	10.00	142.5	21.25
B.	**Receipts / Inflows:** Cash Sales	40×10%=4.00	45×10%=4.50	46×10%=4.60
	Collection from debtors (WN-1)	33.75	36.00	38.25
	Total receipts	37.75	40.50	42.85
C.	**Payments/Outflows:**			
	Payment to creditors (WN1)	29.00	29.00	33.00
	Wages and salaries	3.00	3.00	3.00
	Interim dividend	--	--	2.00
	Machinery purchase – Instalment	0.50	0.50	0.50
	Administration expenses	1.00	1.00	1.00
	Total Payments	33.50	33.50	39.50
D.	**Closing balance / (overdraft)**	**14.25**	**21.25**	**24.60**

Working Note:

Computation of collection from debtors and credit purchases (₹ in lakhs)

	Particulars	Aug	Sep	Oct	Nov	Dec
a.	Total sales	35.00	40.00	40.00	45.00	46.00
b.	Cash sales at 10% of (a)	3.50	4.00	4.00	4.50	4.60
c.	Credit sales (a-b)	31.50	36.00	36.00	40.50	42.40
d.	Collection of debtors:					
	50% in next month	--	15.75	18.00	18.00	20.25
	50% in second month	--	--	15.75	18.00	18.00
e.	COGS [GP Ratio = 20% on sales, So, COGS = 80% of sales, i.e., 80% of (a)]	28.00	32.00	32.00	36.00	36.80
f.	Wages & Salaries (assumed debited to trading a/c)	3.00	3.00	3.00	3.00	3.00
g.	Balance being material consumption cost (e-f) Mat. Pay	25.00	29.00 25	29.00 29	33.00 29	33.80 33

Total Collect

Note: Material consumption cost = Opening stock + Purchases (-) closing stock.

In the absence of information, opening stock = Closing stock. Hence, material consumed = Purchases. Since all purchases are on credit basis only, total purchases (i.e. Material consumed) = Credit purchases.

Alternatively, Wages and salaries can be assumed as other expenses debited to P&L, and hence ignored in the above computations. In such case, COGS = Credit purchases, by following the above analogy.

Question – 4	

The following information relates to SK ltd., a publishing company:

The selling price of a book is ₹15, and sales are made on credit through a book club and invoiced on the last day of the month. Variable costs of production per book are materials (₹5), labour (₹4) and overhead (₹2). +25% = 2.50

The sales manager has forecasted the following volumes:

Month	No. of books	Cost Paym	V. OH
November	1,000	-	-
December	1,000	-	-
January	1,000	5000	1250×2=2500
February	1,250	6250	1500×2=3000
March	1,500	7500	2000×2=4000
April	2,000	10000	
May	1,900	8500	
June	2,200		
July	2,200		
August	2,300		

Customers are expected to pay as follows:

One month after the sale 40%

Two months after the sale 60%

V.OH → Dec.

Nov. ← Prod. — Jan.

Paid → Jan.

The company produces the books two months before they are sold and the creditors for materials are paid two months after production.

Nov. ← Prod — Jan.

V.OH → Pay Dec.

Variable overheads are paid in the month following production and are expected to increase by 25% in April; 75% of wages are paid in the month of production and 25% in the following month. A wage increase of 12.5% will take place on 1st March.

The company is going through a restructuring and will sell one of its freehold properties in May for ₹25,000, but it is also planning to buy a new printing press in May for ₹10,000. Depreciation is currently ₹1,000 per month, and will rise to ₹1,500 after the purchase of the new machine.

The company's corporation tax (of ₹10,000) is due for payment in march. The company presently has a cash balance at bank on 31 December 2020, of ₹1,500.

You are required to prepare a cash budget for the six months from January to June 2021.

Solution

Workings:

1. Sales receipts

Month	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Forecast sales (S)	1,000	1,000	1,000	1,250	1,500	2,000	1,900	2,200
	₹	₹	₹	₹	₹	₹	₹	₹
S × 15	15,000	15,000	15,000	18,750	22,500	30,000	28,500	33,000
Debtors pay:	40%	60%		40%	60%			
1 month 40%	-	6,000	6,000	6,000	7,500	9,000	12,000	11,400
2 month 60%	-	-	9,000	9,000	9,000	11,250	13,500	18,000
Total	-	-	15,000	15,000	16,500	20,250	25,500	29,400

2. Payment for materials

Month	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Qty produced (Q)	1,000 Jan.	1,250 Feb.	1,500 M.	2,000	1,900	2,200	2,200	2,300
	₹	₹	₹	₹	₹	₹	₹	₹
Materials (Q×5)	5,000	6,250	7,500	10,000	9,500	11,000	11,000	11,500
Paid (2 month after)	-	-	5,000	6,250	7,500	10,000	9,500	11,000

3. Variable overheads

Month	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Qty produced (Q)	1,000 Jan.	1,250 Feb.	1,500	2,000	1,900	2,200	2,200	2,300
	₹	₹	₹	₹	₹	₹	₹	₹
Var. Ohs (Q×2)	2,000	2,500	3,000	4,000	3,800	-	-	-
Var. Ohs (Q×2.5)	-	-	-	-	-	5,500	5,500	5,750
Paid one month later	-	2,000	2,500	3,000	4,000	3,800	5,500	5,500

4. Wages payment

Month	Dec	Jan	Feb	Mar	Apr	May	Jun
Qty produced (Q)	1,250	1,500	2,000	1,900	2,200	2,200	2,300
	Feb. ₹	Mar. ₹	₹	₹	₹	₹	₹
Wages (Q×4)	5,000	6,000	8,000	-	-	-	-
Wages (Q × 4.50)	75% -	25% 75% 25%	-	8,550	9,900	9,900	10,350
75% this month	3,750	4,500	6,000	6,412	7,425	7,425	7,762
25% of Pr. month	-	1,250	1,500	2,000	2,138	2,475	2,475
Total	-	5,750	7,500	8,412	9,563	9,900	10,237

Cash Budget

Particulars	Jan	Feb	March	April	May	June
	₹	₹	₹	₹	₹	₹
Receipts:						
Sales receipts	15,000	15,000	16,500	20,250	25,500	29,400
Freehold property	-	-	-	-	25,000	-
	15,000	**15,000**	**16,500**	**20,250**	**50,500**	**29,400**
Payments:						
Materials	5,000	6,250	7,500	10,000	9,500	11,000
Var. Ohs	2,500	3,000	4,000	3,800	5,500	5,500
Wages	5,750	7,500	8,412	9,563	9,900	10,237
Printing press	-	-	-	-	10,000	-
Corporation tax	-	-	10,000	-	-	-
	13,250	**16,750**	**29,921**	**23,363**	**34,900**	**26,737**
Net cash flow	1,750	(1,750)	(13,412)	(3,113)	15,600	2,663
Op. Bal.	1,500	3,250	1,500	(11,912)	(15,025)	575
Closing bal.	3,250	1,500	(11,912)	(15,025)	575	3,238

Question – 5	

Based on the following information prepare a cash budget for SK Ltd:

	1st Qtr (₹)	2nd Qtr (₹)	3rd Qtr (₹)	4th Qtr (₹)
Opening cash balance	10,000	—	—	—
Collections from customers	1,25,000	1,50,000	1,60,000	2,21,000
Payments:				
Purchase of materials	20,000	35,000	35,000	54,200
Other expenses	25,000	20,000	20,000	17,000
Salary and wages	90,000	95,000	95,000	1,09,200
Income tax	5,000	—	—	—
Purchase of machinery	—	—	—	20,000

The company desired to maintain a cash balance of ₹ 15,000 at the end of each quarter. Cash can be borrowed or repaid in multiples of ₹ 500 at an interest of 10% per annum. Management does not want to borrow cash more than what is necessary and wants to repay as early as possible. In any event, loans cannot be extended beyond four quarters. Interest is computed and paid when the principal is repaid. Assume that borrowings take place at the beginning and repayments are made at the end of the quarters.

Solution

Cash Budget

Particulars	Quarter-1	Quarter-2	Quarter-3	Quarter-4
Opening Balance (A)	**10,000**	**15,000**	**15,000**	**15,325**
Collection from customers (B)	**1,25,000**	**1,50,000**	**1,60,000**	**2,21,000**
Payments:				
Purchase of material	20,000	35,000	35,000	54,200
Other expenses	25,000	20,000	20,000	17,000
Salary	90,000	95,000	95,000	1,09,200
Income Tax	5,000	-	-	-
Purchase of Machinery	-	-	-	20,000
Total Payments (C)	**1,40,000**	**1,50,000**	**1,50,000**	**2,00,400**
Surplus\(Deficit) (A + B – C)	**(5,000)**	**15,000**	**25,000**	**35,925**
Add: Borrowing	20,000	-	-	-
Less: Principal Repayment		-	9,000	11,000
Less: Interest payment	-	-	675	1,100
Closing Balance	**15,000**	**15,000**	**15,325**	**23,825**

Working notes:

(1) Since there was deficit of ₹ 5,000 in Q-1, thus borrowing will be done by company of ₹ 20,000 so that the closing balance stands at ₹ 15,000.

(2) In Q-2, there is neither any surplus nor and deficit as compared to the required minimum balance. So neither any borrowing nor any repayment can be done.

(3) In Q-3, the company has ₹ 10,000 in excess than the minimum required closing balance. This can be used to repay the amount borrowed in Q-1.

Let principal to be repay in Q-1 = y

∴ Interest = y × 10% × (3/4) = 0.075y

y + 0.075y = 10,000

y = ₹ 9,302

∴ Principal repayment in multiples of ₹ 500 that can be done will be ₹ 9,000

Interest on amount repaid = 9,000 × 10% × (3/4) = ₹ 675

(4) In Q-4, the company has ₹ 20,925 in excess than the minimum required closing balance. This will be used to repay the balance amount of debt outstanding i.e. ₹ 11,000. Interest on this will be ₹ 1,100 (11,000 × 10%).

Question – 6

You are given below the Profit & Loss Accounts for two years for a company:

Profit & Loss Account

Particulars	Year 1(₹)	Year 2(₹)	Particulars	Year 1(₹)	Year 2(₹)
To Opening Stock	40,00,000	50,00,000	By Sales	4,00,00,000	5,00,00,000
To Raw Materials 210₹	1,50,00,000	2,00,00,000	By Closing Stock	50,00,000	75,00,000
To Stores	50,00,000	60,00,000	By Misc. Income	5,00,000	5,00,000
To Manufacturing Exp.	50,00,000	80,00,000			
To Other Expenses	50,00,000	50,00,000			
To Depreciation	50,00,000	50,00,000			
To Net Profit	65,00,000	90,00,000			
	4,55,00,000	5,80,00,000		4,55,00,000	5,80,00,000

48.3
600₹

Sales are expected to be ₹ 6,00,00,000 in year 3. Other expenses will increase by ₹ 25,00,000 besides other charges. Only raw materials are in stock. Assume sales and purchases are in cash terms and the closing stock is expected to go up by the same amount as between year 1 and 2. You may assume that no dividend is being paid. The Company can use 75% of the cash generated to service a loan. Compute how much cash from operations will be available in year 3 for the purpose? Ignore income tax.

Solution

Projected Profit & Loss Account for Year 3

Particulars	Year 3 (₹)	Particulars	Year 3 (₹)
To Material Consumed (w.n. – 1)	2,10,00,000	By Sales	6,00,00,000
To Stores (w.n. – 2)	72,00,000	By Misc. Income	5,00,000
To Manufacturing Exp. (w.n. – 3)	96,00,000		
To Other Expenses	75,00,000		
To Depreciation	50,00,000		
To Net Profit	1,02,00,000		
	6,05,00,000		6,05,00,000

Cash Flow

Particulars	Amount
Profit	1,02,00,000
Add: Depreciation	50,00,000
Less: cash required for increase in stock	(25,00,000)
Net cash inflow	**1,27,00,000**

Amount available for servicing the loan = 75% × 1,27,00,000 = ₹ 63,50,000

Working Notes

(1) Material consumed in year 2 = 50,00,000 + 2,00,00,000 – 75,00,000 = ₹ 1,75,00,000

Material consumed in year 2 as % of sales = $\frac{1,75,00,000}{5,00,00,000} \times 100$ = 35% of sales

Material consumption in year 3 = 6,00,00,000 × 35% = ₹ 2,10,00,000

(2) Stores as % of sales in year 2 = $\frac{60,00,000}{5,00,00,000} \times 100$ = 12% of sales

Stores in year 3 = 6,00,00,000 × 12% = ₹ 72,00,000

(3) Manufacturing expenses as % of sales in year 2 = $\frac{80,00,000}{5,00,00,000} \times 100$ = 16% of sales

Manufacturing expenses in year 3 = 6,00,00,000 × 16% = ₹ 96,00,000